

On images, preimages and different notions related to APN mappings

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Abstract

Recently, there has been an increasing interest in studying the behavior of vectorial functions $F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ on affine subspaces of $\mathbb{F}_2^n$. Kolomeec and Bykov [KB24] studied the images of the inverse/power functions on k -dimensional flats of $\mathbb{F}_p^n$ aimed to understand which of them remain flats. Those that are not mapped to flats are called *broken*. In this paper, we specify exactly the number of unbroken 2-dimensional flats for certain APN mappings, and moreover demonstrate that this property can be used to derive a lower bound on the vectorial nonlinearity of APN functions. We also introduce the concept of *strong breaking* of k -dimensional affine flats which implies the standard breaking property. Finally, we study the restrictions on the preimage sets of APN functions, as well as the bounds on the maximal dimension of affine flats that can occur in their image space. In particular, the image space of an APN function cannot be a hyperplane. However, we provide examples of APN functions in odd dimension which contain a hyperplane in their image set.

Key Words : Vectorial functions, APN functions, differential uniformity, image set, affine subspaces, hyperplanes.

1 Introduction

We first recall some basic concepts and notation. Throughout the paper, we identify the finite field $\mathbb{F}_{2^n}$ with the vector space $\mathbb{F}_2^n$, by fixing a basis

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of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$. The *Hamming weight* of $x = (x_1, \dots, x_n) \in \mathbb{F}_2^n$ is defined as $w_H(x) = |\{1 \leq i \leq n \mid x_i \neq 0\}|$, and for $x, x' \in \mathbb{F}_2^n$, the *Hamming distance between x and x'* is defined as $d_H(x, x') = w_H(x + x')$. A function $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is *k-to-1* if every element of $\text{Im}(F)$ has exactly k preimages, and it is *almost k-to-1* if exactly one image element has one preimage and all others have k preimages. For a polynomial F over $\mathbb{F}_{2^n}$ and $a \in \mathbb{F}_{2^n}^*$, the *first-order derivative of F at a* is the polynomial $F(x + a) + F(x)$. Given a polynomial F and field elements $a \in \mathbb{F}_{2^n}^*$, $b \in \mathbb{F}_{2^n}$, we define

$$\delta_F(a, b) = |\{x \in \mathbb{F}_{2^n} : F(x + a) + F(x) = b\}|.$$

Clearly, $\delta_F(a, b)$ must be even (since, if x_0 is a solution of the above equation, so is $x_0 + a$). The *differential uniformity* of F is defined as

$$\delta_F = \max_{a \in \mathbb{F}_{2^n}^*, b \in \mathbb{F}_{2^n}} \delta_F(a, b).$$

The polynomial F is called *almost perfect nonlinear* (APN), if its differential uniformity is $\delta_F = 2$, which is the smallest possible. Equivalently, F is APN if and only if the first-order derivative $F(x + a) + F(x)$ is a 2-to-1 mapping for each $a \in \mathbb{F}_{2^n}^*$. The *absolute trace function* over $\mathbb{F}_{2^n}$ is defined as $\text{Tr}(x) = x + x^2 + \dots + x^{2^{n-1}}$.

A *k-dimensional flat* (or *k-flat*) is a k -dimensional affine subspace of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$. When dimension is not relevant or when it is obvious from the context, simply *flat* will be used. A *hyperplane* of $\mathbb{F}_{2^n}$ is an $(n - 1)$ -dimensional subspace of $\mathbb{F}_{2^n}$ over $\mathbb{F}_2$. Every hyperplane of $\mathbb{F}_{2^n}$ can be written in the form $H_a := \{y \in \mathbb{F}_{2^n} \mid \text{Tr}(ay) = 0\}$, for a unique $a \in \mathbb{F}_{2^n}^*$. Given any subset $S \subseteq \mathbb{F}_{2^n}$, we reserve the notation $\overline{S}$ to denote $\mathbb{F}_{2^n} \setminus S$. In particular, $\overline{H}_a$ is an $(n - 1)$ -dimensional flat. For $n \geq 2$, define the set of all 2-dimensional flats in $\mathbb{F}_{2^n}$ as

$$\mathcal{B}_n = \{\{x_1, x_2, x_3, x_4\} \mid \sum_{i=1}^4 x_i = 0 \text{ and } x_i \in \mathbb{F}_{2^n} \text{ are distinct}\}.$$

Conventionally, each subset in $\mathcal{B}_n$ is called a *block*. The classical *Steiner quadruple system* is a pair $(\mathbb{F}_{2^n}, \mathcal{B}_n)$, so that each 3-subset of $\mathbb{F}_{2^n}$ is contained in exactly one block of $\mathcal{B}_n$. A function $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is APN if and only if for each $\{x_1, x_2, x_3, x_4\} \in \mathcal{B}_n$,

$$F(x_1) + F(x_2) + F(x_3) + F(x_4) \neq 0.$$

Namely, the summation of F over each 2-dimensional flat is non-vanishing. For a function $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$, define the set of *vanishing flats* as [LMP+20]

$$\mathcal{VB}_{n,F} = \{\{x_1, x_2, x_3, x_4\} \in \mathcal{B}_n \mid F(x_1) + F(x_2) + F(x_3) + F(x_4) = 0\}.$$

In this work, we will consider recently introduced notions related to APN mappings, such as the breaking property [KB24], *k*th-order-non-affine and

k th-order-sum-free property introduced by Carlet [Car25]. For certain APN mappings, we derive the exact number of unbroken 2-dimensional flats, see Theorem 2.2. Moreover, to further refine the breaking property of non-injective APN mappings we introduce the *strong breaking* property which implies the standard breaking property (see Proposition 2.5), but the converse is not true. On the other hand, the standard breaking property of order k (and therefore its strong version as well) implies the property of k -strongly non-normality (the concept introduced by Carlet in [Car25]). Another application of the breaking property is a lower bound on the vectorial nonlinearity given in Theorem 3.1, which is unfortunately not as strong as a recent result of Nagy [Nag25]. Finally, we analyze the image set of a non-permutation APN mapping by observing that it cannot be a hyperplane. On the other hand, the image set of such an APN function may actually contain a hyperplane, as indicated by some examples in odd dimensions. In addition, the results of our computer simulations (which will be included in the long version of this paper) motivate the investigation of the structure of the image sets of quadratic APN mappings in even dimensions and whether a hyperplane can be contained in their image sets.

2 Different notions related to APN mappings

In this section, we deduce certain results regarding the so-called breaking property of APN mappings (see Definition 2.1) and we introduce another relevant notion that will be referred to as *strong breaking*. Moreover, the relationship between these two properties and the vanishing properties of APN mappings will be addressed.

2.1 Strong breaking of k -dimensional flats

Certain characterization of APN functions can be deduced by considering the images of k -dimensional flats. More precisely, the following definition was used in [KB24].

Definition 2.1. *We say that a mapping $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ breaks a flat L of $\mathbb{F}_{2^n}$ if $F(L)$ is not a flat of $\mathbb{F}_{2^n}$.*

A study of the properties of breaking affine subspaces was considered recently in [KB24]. In particular, the inverse function $F(x) = x^{-1}$ maps the minimum number of affine subspaces to affine subspaces among all invertible power functions. In addition, a sufficient condition was provided so that a function $A(x^{-1}) + b$ has no invariant affine subspaces U of cardinality $2 < |U| < p^n$ for an invertible linear transformation $A: \mathbb{F}_{p^n} \rightarrow \mathbb{F}_{p^n}$ and $b \in \mathbb{F}_{p^n}$.

In the sequel, the preimage of an element $a \in \mathbb{F}_{2^n}$ under F will be denoted by $F^{-1}[a]$, i.e.,

$$F^{-1}[a] = \{x \in \mathbb{F}_{2^n} : F(x) = a\}.$$

For $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$, define the set of unbroken 2-dimensional flats with respect to F as

$$\mathcal{UB}_{n,F} = \{L \in \mathcal{B}_n \mid F(L) \text{ is a flat}\}.$$

Theorem 2.2. *Let $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be an APN function. Consider the set*

$$\mathcal{P} := \{F^{-1}[a] \mid a \in \mathbb{F}_{2^n}, |F^{-1}[a]| \geq 3\}.$$

If $\mathcal{P} \neq \emptyset$, then the number of unbroken 2-dimensional flats is

$$|\mathcal{UB}_{n,F}| = \sum_{P \in \mathcal{P}} \binom{|P|}{3}.$$

Otherwise, $|\mathcal{UB}_{n,F}| = 0$. In particular, if F is almost k -to-one (thus k divides $2^n - 1$) with $k \geq 3$ then the number of unbroken 2-dimensional flats is

$$\frac{\binom{k}{3}(2^n - 1)}{k} = \frac{(2^n - 1)(k - 1)(k - 2)}{6}.$$

Proof. Let $L = \{x_1, x_2, x_3, x_4\}$ be a 2-dimensional flat. Note that the restriction $F|_L$ of f to L is neither constant nor two-to-one since F is APN (otherwise $\sum_{x \in L} F(x)$ would be 0). Moreover, $L \notin \mathcal{UB}_{n,F}$ whenever $|F(L)| = 3$. If $F|_L$ is one-to-one on L , then F breaks L as F is APN, whereas if $F|_L$ is almost 3-to-one on L then L is mapped to a 1-dimensional flat. To see the latter, assume w.l.o.g. that $\{x_1, x_2, x_3\} \subseteq F^{-1}[a]$ and $F(x_4) = b$ for some $a, b \in \mathbb{F}_{2^n}$ with $a \neq b$. This means that $F(L) = \{a, a + (a + b)\}$. We then conclude that the only unbroken flats L must satisfy that $F|_L$ is almost 3-to-one. Moreover, every three different elements in a fiber of F determine such a flat. Thus, if the set $\mathcal{P}$ of fibers with more than 3 elements is nonempty, then the total number of unbroken flats is $\sum_{P \in \mathcal{P}} \binom{|P|}{3}$. Otherwise, every 2-dimensional flat is broken. $\square$

Example 2.3. *Let $F : \mathbb{F}_{2^6} \rightarrow \mathbb{F}_{2^6}$ be the Gold function defined by $F(x) = x^3$. Since F is 3-to-1, Theorem 2.2 implies that $|\mathcal{UB}_{6,F}| = 21$; in other words, exactly 21 two-dimensional flats are unbroken with respect to F . Using the vectorial representation, these unbroken flats are*

$$\begin{aligned} \mathcal{UB}_{6,F} = \{ & \{38, 0, 24, 62\}, \{21, 46, 0, 59\}, \{29, 0, 16, 13\}, \\ & \{40, 0, 49, 25\}, \{51, 0, 54, 5\}, \{27, 47, 0, 52\}, \\ & \{19, 0, 31, 12\}, \{55, 0, 61, 10\}, \{35, 8, 0, 43\}, \\ & \{23, 0, 39, 48\}, \{22, 41, 63, 0\}, \{50, 57, 0, 11\}, \end{aligned}$$

$$\begin{aligned} &\{14, 15, 0, 1\}, \{30, 0, 2, 28\}, \{3, 0, 18, 17\}, \\ &\{37, 0, 9, 44\}, \{56, 60, 0, 4\}, \{0, 33, 53, 20\} \\ &\{0, 32, 58, 26\}, \{7, 0, 42, 45\}, \{36, 0, 6, 34\}, \end{aligned}$$

where each flat $\{a, b, c, d\}$ is written via the integer values of elements of $\mathbb{F}_2^6$. For example, the flat $\{36, 0, 6, 34\}$ corresponds to

$$\{(1, 0, 0, 1, 0, 0), (0, 0, 0, 0, 0, 0), (0, 0, 0, 1, 1, 0), (1, 0, 0, 0, 1, 0)\},$$

where each integer is converted to its binary vector representation in $\mathbb{F}_2^6$.

For non-bijective APN mappings, there are always vanishing images (e.g., when F is a 2-to-1, then for every $a \in \mathbb{F}_{2^n}$ there is a unique $b \in \mathbb{F}_{2^n}$ such that $a \neq b$ and $F(a) + F(b) = 0$) that may hide the proper structure of the image set. In order to better understand the image sets in relation with the breaking property, we introduce the following definition.

Definition 2.4. Let $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be a function and L be a flat of $\mathbb{F}_{2^n}$. We say that F strongly breaks L if

$$\sum_{y \in F(L)} y \neq 0,$$

where $F(L)$ denotes the image of L under F .

For any set $X \subseteq \mathbb{F}_{2^n}$, $\sum_{x \in X} x \neq 0$ implies that X is not a flat. Therefore, we get the following.

Proposition 2.5. Let $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be a function and L be any flat such that $|F(L)| > 2$. If F strongly breaks L , then F breaks L .

In general, the converse of Proposition 2.5 is not true—not even for APN permutations, e.g. it can be verified that Dillon’s permutation [BDMW10] breaks all 3-flats, but there are 184 3-flats which are not strongly broken.

2.2 k -strongly nonnormal functions

In a recent article, Carlet introduced [Car25] two notions related to properties of vectorial mappings $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$, referred to as (n, m) -functions.

Definition 2.6. Let $2 \leq k \leq n$ and m be positive integers. An (n, m) -function F is called k -strongly non-normal (resp. k th-order sum-free) if, for every k -flat A of $\mathbb{F}_2^n$, the restriction of F to A is not an affine function (resp. the sum $\sum_{x \in A} F(x)$ is nonzero).

It was noticed in [Car25], that the k th-order sum-freeness implies k -strong non-normality, since the sum of the values taken by an affine function over an affine space of dimension at least 2 equals 0. Moreover, k th-order

sum-freedom is a strong property and k -strong non-normality is a much weaker one (contrary to what its name evokes).

There is then a connection between third-order sum-freedom and the so-called *D-property*, due to Dillon, saying that the union of the image sets of all such mappings ψ_E , when E ranges over the set of all affine planes, covers $\mathbb{F}_2^m \setminus \{0\}$. Also, all APN functions (which exist for every n) are k -strongly non-normal for every $2 \leq k \leq n$ [Car25]. In fact, an (n, m) -function F is k th-order sum-free if, and only if, the restriction of F to any k -dimensional affine space, viewed as a k -variable function through the choice of a basis of the underlying vector space, has algebraic degree k .

Remark 2.7. *The property of a function being k -strongly non-normal is a CCZ-invariant (Proposition 6, [Car25]), and the property of breaking is not invariant under CCZ equivalence. For example, for APN permutations it may hold that vanishing and breaking coincide for 2-dim. flats, i.e. there are no 2-dim. unbroken flats, whereas a CCZ-equivalent APN function might have some unbroken flats.*

If there exists a k -flat A such that the restriction of F to A is affine, then the image of A must be a flat. Thus we have the following proposition.

Proposition 2.8. *If $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^m}$ breaks all k -dimensional flats, then F is k -strongly non-normal.*

Proof. Let $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^m}$ be a function that breaks all k -flats, and suppose that F is not k -strongly non-normal. Then, there exists a k -flat $A \subseteq \mathbb{F}_{2^n}$ for which the restriction $F|_A$ is affine, i.e.,

$$F(x) = l(x) + b, \quad \text{for all } x \in A,$$

where $l: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^m}$ is a linear function and $b \in \mathbb{F}_{2^m}$. Since A is an affine subspace, we may write it in the form $A = U + a$, where $a \in \mathbb{F}_{2^n}$ and U is a k -dimensional linear subspace of $\mathbb{F}_{2^n}$. Then,

$$\begin{aligned} F(A) &= \{ F(u + a) : u \in U \} \\ &= \{ l(u + a) + b : u \in U \} \\ &= \{ l(u) + l(a) + b : u \in U \} \\ &= \{ v + c : v \in l(U) \} \\ &= V + c, \end{aligned}$$

where $V = l(U)$ is a linear subspace of $\mathbb{F}_{2^m}$ and $c = l(a) + b \in \mathbb{F}_{2^m}$. Thus $F(A)$ is an affine subspace of $\mathbb{F}_{2^m}$, contradicting the assumption that F breaks all k -flats. $\square$

3 Lower bound on vectorial nonlinearity via breaking property

In this section, we provide a lower bound on vectorial nonlinearity introduced in [LMC17] by employing the cardinality of broken/unbroken 2-dimensional flats. More precisely, the vectorial nonlinearity of $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ is

$$\mathcal{NL}_V F = \min_{\Phi \in \mathcal{A}_{n,m}} d_H(F, \Phi), \quad (3.1)$$

where $\mathcal{A}_{n,m}$ denotes the set of affine functions from $\mathbb{F}_2^n$ to $\mathbb{F}_2^m$, and $d_H(F, \Phi) = |\{x \in \mathbb{F}_2^n : F(x) \neq \Phi(x)\}|$. The main idea is to simply use the estimate on the number of broken/unbroken 2-dimensional flats, which then gives the lower bound.

Theorem 3.1. *Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ be a mapping for which all 2-dimensional flats are broken. Then, the vectorial nonlinearity is lower bounded by 2^{n-2} . Moreover, if the number of unbroken 2-dimensional flats is $|\mathcal{UB}_{n,F}| = w$, then $\mathcal{NL}_V F \geq 2^{n-2} - w$.*

Proof. We will represent $\mathbb{F}_2^n$ as a union of 2-dimensional flats, thus $\mathbb{F}_2^n = \bigcup_{i=1}^{2^{n-2}} S_i$, where S_i are disjoint 2-dimensional flats. Then, for any $G \in \mathcal{A}_{n,n}$, its restriction $G|_{S_i}$ to S_i is also affine, and therefore

$$\mathcal{NL}_V F = \min_{G \in \mathcal{A}_{n,n}} \left(\sum_{i=1}^{2^{n-2}} d_H(F|_{S_i}, G|_{S_i}) \right) \geq 2^{n-2} - w,$$

since if S_i is broken, then $F|_{S_i}$ is not affine, and hence $d_H(F|_{S_i}, G|_{S_i}) \geq 1$. So if $|\mathcal{UB}_{n,F}| = w$, then at least $2^{n-2} - w$ flats are broken. In particular, when $|\mathcal{UB}_{n,F}| = 0$ then $\mathcal{NL}_V F \geq 2^{n-2}$. $\square$

Corollary 3.2. *Let $F(x) = x^{-1}$ be the inverse function over $\mathbb{F}_{2^n}$, with n odd. Then, $\mathcal{NL}_V F \geq 2^{n-2}$.*

Proof. Since the inverse function breaks all 2-dimensional flats [KB24], the result follows from Theorem 3.1. $\square$

Remark 3.3. *The problem of computing vectorial nonlinearity is considered difficult as mentioned in [LMC17] (see also [Nag25]). Carlet [Car22], in a private communication with G. Nagy, suggested a lower bound on vectorial nonlinearity so that $\mathcal{NL}_V F \geq 2^n - \sqrt{2^n + \delta_F(2^n - 1)}$, which was recently improved by Nagy $\mathcal{NL}_V F \geq 2^n - \sqrt{\delta_F} 2^{n/2} - 1/2$.*

The bound in Theorem 3.1 can be possibly improved by employing further the structure of affine and APN functions in terms of their decomposition to 2-dimensional affine spaces, especially for quadratic APN functions.

Open Problem 3.4. *Provide a tight lower bound on the vectorial nonlinearity of quadratic APN functions.*

4 Preimage sets of certain APN mappings

The main goal of this section is to investigate the exact properties of APN mappings with respect to their preimage sets.

4.1 Preimage intersections

Using a similar terminology as in [KKK23], for any function $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ and a k -dimensional flat A of $\mathbb{F}_{2^n}$, we define the numbers $\lambda_{A,i}$, $0 \leq i \leq 2^k$ to be the number of preimages of F that intersect A in i elements, i.e.,

$$\lambda_{A,i} := |\{b \in \mathbb{F}_{2^n} : |F^{-1}[b] \cap A| = i\}|.$$

It clearly holds

$$\sum_{i=1}^{2^k} i \lambda_{A,i} = 2^k. \quad (4.1)$$

Moreover, if the image of A is again a flat, say, l -dimensional, then

$$\sum_{i=1}^{2^k} \lambda_{A,i} = 2^l. \quad (4.2)$$

These two equations constrain the possible values that the λ 's can take.

Proposition 4.1. *Let $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be a function such that all its preimage sets have size at most 3. For any k -dimensional flat A whose image is an l -dimensional flat, with $l \geq 2$, the numbers $\lambda_{A,1}$, $\lambda_{A,2}$ and $\lambda_{A,3}$ can have exactly one of the following patterns $(\lambda_{A,1}, \lambda_{A,2}, \lambda_{A,3})$ modulo 4: $(0, 0, 0)$, $(2, 0, 2)$, $(1, 2, 1)$, or $(-1, 2, -1)$.*

Proof. Since $k, l \geq 2$, from equations (4.1) and (4.2) it follows that

$$\lambda_{A,1} + 2\lambda_{A,2} + 3\lambda_{A,3} \equiv 0 \pmod{4} \quad (4.3)$$

and

$$\lambda_{A,1} + \lambda_{A,2} + \lambda_{A,3} \equiv 0 \pmod{4}. \quad (4.4)$$

We consider the following three cases. If $\lambda_{A,1} \equiv 0 \pmod{4}$, then from equations (4.3) and (4.4) it follows

$$2\lambda_{A,2} + 3\lambda_{A,3} \equiv 0 \pmod{4}$$

$$\lambda_{A,2} + \lambda_{A,3} \equiv 0 \pmod{4}.$$

Adding up the equations above gives $3\lambda_{A,2} \equiv 0 \pmod{4}$, i.e. $\lambda_{A,2} \equiv 0 \pmod{4}$. Lastly, from (4.4) it follows that then $\lambda_{A,3} \equiv 0 \pmod{4}$. If $\lambda_{A,1} \equiv 2 \pmod{4}$, then

$$2\lambda_{A,2} + 3\lambda_{A,3} \equiv 2 \pmod{4}$$

$$\lambda_{A,2} + \lambda_{A,3} \equiv 2 \pmod{4}.$$

Similarly as before, adding up the two equations above we get $3\lambda_{A,2} \equiv 0 \pmod{4}$, i.e. $\lambda_{A,2} \equiv 0 \pmod{4}$. Then, from (4.4) it follows that also $\lambda_{A,3} \equiv 2 \pmod{4}$. If $\lambda_{A,1} \equiv \pm 1 \pmod{4}$, then from equations (4.3) and (4.4) it follows

$$2\lambda_{A,2} + 3\lambda_{A,3} \equiv \mp 1 \pmod{4}$$

$$\lambda_{A,2} + \lambda_{A,3} \equiv \mp 1 \pmod{4}.$$

Multiplying the second equation by 2, and subtracting it from the first equation we get $\lambda_{A,3} \equiv \pm 1 \pmod{4}$. Lastly, from (4.4) it follows that i.e. $\lambda_{A,2} \equiv 2 \pmod{4}$. $\square$

From the previous results, we see that if $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is a 2-to-1 function, then the only possibly unbroken k -flats are such that their image is a $(k-1)$ -flat (the restriction to this flat is 2-to-1) or a k -flat again (i.e. the restriction to this flat is injective). Indeed, Equations (4.1) and (4.2) imply $\lambda_{A,1} = 2^{l+1} - 2^k$ and $\lambda_{A,2} = 2^k - 2^l$. Since $\lambda_{A,1} \geq 0$ then $2^{l+1} \geq 2^k$, equivalently, $l \geq k-1$.

Proposition 4.2. *Let $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be a 2-to-1 function that is k th-order-sum-free. Then F breaks all k -flats.*

Proposition 4.3. *For a function $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ whose all preimage sets have size at most 3, the only possible unbroken 3-dimensional flats satisfy*

$$(\lambda_{A,1}, \lambda_{A,2}, \lambda_{A,3}) \in \{(8, 0, 0), (0, 4, 0), (2, 0, 2), (1, 2, 1)\}.$$

5 On the maximal dimension of flats in the image space

There are a few results on the image sets of APN functions which can be found in the paper by Kyureghyan et al. [KKK23], [BIK23] and the references therein. One of the most important results of [KKK23] states that, for even n , we have that $|\text{Im}(F)| \geq (2^n + 2)/3$ when F is APN. It moreover holds that the image set of any APN function cannot be a hyperplane. The following proposition establishes this folklore result.

Proposition 5.1. *Let $F: \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ be an APN function, with $n > 2$. Then $\text{Im}(F)$ can not be a hyperplane.*

Proof. Suppose that $\text{Im}(F)$ is a hyperplane, so $\dim(\text{Im}(F)) = n-1$. Then F can be viewed as an $(n, n-1)$ -function. Since F is APN, we have that its differential uniformity is $\delta_F = 2 = 2^{n-(n-1)}$. This equality implies that F is bent [Car21, p. 135]. However, bent functions must have codomain dimension at most $\frac{n}{2}$. Since $n-1 > \frac{n}{2}$ for $n > 2$, this leads to a contradiction. $\square$

Given a quadratic non-bijective APN, it holds $|\text{Im}(F)| \leq 2^{n-1} - 2^{\frac{n-1}{2}}$ when n is odd, by the results in [KKK23]. Using that l hyperplanes intersect in 2^{n-l} elements, we can show that, if there are l hyperplanes in the image of F , then

$$|\text{Im}(F)| \geq 2^{n-1} + 2^{n-2} + \dots + 2^{n-l} = 2^n - 2^{n-l},$$

which implies that $l \leq \frac{n+1}{2}$. Moreover, this number can be positive, as illustrated by the following two examples.

Example 5.2. Let $F : \mathbb{F}_{2^5} \rightarrow \mathbb{F}_{2^5}$ be defined as $F(x) = x^{-1} + \text{Tr}(x)$ and $a = \omega^4 + \omega$, where ω is a root of the primitive polynomial $X^5 + X^2 + 1 \in \mathbb{F}_2[x]$. Then, F is a non-bijective APN function with algebraic degree 4, whose image contains the hyperplane $H_a = \{x \in \mathbb{F}_{2^5} \mid \text{Tr}(ax) = 0\}$.

Example 5.3. Let $F : \mathbb{F}_{2^9} \rightarrow \mathbb{F}_{2^9}$ be defined as

$$F(x) = x^{144} + x^{130} + x^{72} + x^{65} + x^{18} + x^9 + x^3$$

and $a = \omega^6 + \omega$, where ω is a root of the polynomial $X^9 + X^4 + 1 \in \mathbb{F}_2[x]$. Then, F is a non-bijective quadratic APN function whose image contains the hyperplane $H_a = \{x \in \mathbb{F}_{2^9} \mid \text{Tr}(ax) = 0\}$.

Based on the computer simulations which were performed for quadratic APN non-permutations over $\mathbb{F}_2^n$, where n is even, we pose the following open problem.

Open Problem 5.4. Can the image space of a non-bijective quadratic APN function $F : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$, where n is even, contain a hyperplane (or its complement)?

6 Conclusions

In this article, we have provided several results concerning the recently introduced notions related to APN mappings. In particular, we specify exactly the number of unbroken 2-dimensional flats for certain APN functions. Moreover, using this property, we derived a lower bound on the vectorial nonlinearity of APN functions. We also introduced the concept of *strong breaking* of k -dimensional flats, as a refinement of the breaking property. Finally, we studied the restrictions on the preimage of APN functions, as well as the bounds on the maximal dimension of flats that can occur in their image space. It has been shown that the image space of an APN function cannot be a hyperplane, and we provided examples of APN functions in odd dimension which contain a hyperplane in their image set. We suggest two interesting research tasks about quadratic APN functions, namely, provide a tight lower bound on their vectorial nonlinearity, and investigate whether non-bijective quadratic APN mappings in even dimension can contain a hyperplane in their image.

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